# Indian Institute of Technology, Ropar

# Design Lab Project Report

**Project Title -**

## 3-Axis orientation stabilization platform for vehicle-mounted cameras and LiDAR sensors


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# 1 Abstract

This project presents the design and development of a 3-axis orientation stabilization platform intended for maintaining stable mounting of cameras and LiDAR sensors on vehicles operating in uneven and rough terrain environments. The system addresses the challenge of orientation instability caused by vibrations, sudden accelerations, and multi-directional disturbances that degrade the accuracy of onboard sensing systems in mobile robotics and autonomous platforms.

The proposed system is built around an ESP32 microcontroller as the central processing unit, interfaced with an MPU6050 sensor for real-time acquisition of angular velocity and acceleration data. Orientation estimation is performed for roll, pitch, and yaw axes, where roll and pitch are computed using accelerometer-based trigonometric relations, while yaw is obtained through gyroscope-based angular integration.

A closed-loop Proportional–Integral–Derivative (PID) control algorithm is implemented to continuously minimize the deviation between desired and measured orientation. The control output is mapped to three MG995 high-torque servos through a PCA9685 PWM driver module, enabling real-time mechanical correction of the platform's orientation. The system is designed using a single-platform servo linkage mechanism, allowing coordinated multi-axis stabilization.

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# 2 Objectives

- To design and develop a 3-axis stabilization platform capable of maintaining orientation in roll, pitch, and yaw.
- To acquire real-time motion data using the MPU6050 sensor and compute angular displacement.
- To implement a closed-loop control system for minimizing orientation error.
- To actuate servo motors using ESP32 and PCA9685 for precise mechanical correction.
- To evaluate system performance under external disturbances such as manual shaking and tilt.

# 3 Achievements

- Successful design and implementation of a functional 3-axis stabilization platform.
- Real-time acquisition and processing of sensor data using ESP32.
- Effective reduction of angular disturbances using proportional control.
- Demonstration of stable platform behavior under manually induced disturbances.
- Development of MATLAB-based simulation model for system validation.

## 4 Future Scope

- Implementation of advanced sensor fusion techniques such as Kalman filtering for improved accuracy.
- Integration of magnetometer for accurate yaw estimation and drift reduction.
- Development of adaptive or self-tuning PID controllers for varying terrain conditions.
- Mechanical redesign using a gimbal-based structure to reduce axis coupling.
- Integration with real-world autonomous vehicle or robotic platforms.
- Use of higher precision actuators for improved response and stability.

## 5 Methodology

The system follows a closed-loop control architecture:

$$\text{Sensor Data} \rightarrow \text{Angle Estimation} \rightarrow \text{Error Calculation} \rightarrow \text{PID Control} \rightarrow \text{Servo Actuation}$$

### 5.1 Orientation Estimation

Roll and pitch angles are calculated using accelerometer data:

$$\phi = \tan^{-1}\left(\frac{a_y}{a_z}\right)$$

$$\theta = \tan^{-1}\left(\frac{-a_x}{\sqrt{a_y^2 + a_z^2}}\right)$$

Yaw is estimated using gyroscope integration:

$$\psi(t) = \psi(t-1) + \omega_z \cdot dt$$

### 5.2 Error Computation

$$e(t) = \theta_{desired} - \theta_{measured}$$

For all axes:

$$e_\phi, \quad e_\theta, \quad e_\psi$$

### 5.3 PID Controller

The control signal is given by:

$$u(t) = K_p e(t) + K_i \int e(t)dt + K_d \frac{de(t)}{dt}$$

Discrete form:

$$u[k] = K_p e[k] + K_i \sum e[k]dt + K_d \frac{e[k] - e[k-1]}{dt}$$

### 5.4 System Modelling of MG995 Servomotor

Based on system identification results, the MG995 servomotor can be more accurately modeled as a second-order linear time-invariant system. The transfer function relating the input reference signal to the output angular position is expressed as:

$$\frac{\Theta_{out}(s)}{\Theta_{ref}(s)} = \frac{224.8}{s^2 + 22.33s + 225.4} \tag{1}$$

Where:

- $\Theta_{ref}(s)$ is the Laplace transform of the input reference angle
- $\Theta_{out}(s)$ is the Laplace transform of the output shaft angle

### 5.5 Standard Second-Order System Representation

The general form of a second-order system is:

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \tag{2}$$

Comparing with the identified transfer function:

$$s^2 + 22.33s + 225.4 \tag{3}$$

We obtain:

$$\omega_n^2 = 225.4 \quad \Rightarrow \quad \omega_n = \sqrt{225.4} \approx 15.01\,\text{rad/s} \tag{4}$$

$$2\zeta\omega_n = 22.33 \tag{5}$$

Solving for damping ratio:

$$\zeta = \frac{22.33}{2 \times 15.01} \approx 0.74 \tag{6}$$

## 5.6 System Interpretation

From the above parameters:

- The natural frequency $\omega_n \approx 15.01$ rad/s indicates a fast system response.
- The damping ratio $\zeta \approx 0.74$ suggests an underdamped system with limited oscillations.

This implies that the MG995 servomotor exhibits:

- Rapid response to input commands
- Moderate overshoot
- Stable convergence to the desired position

## 5.7 Time-Domain Characteristics

The performance of a second-order system can be approximated using:

$$t_s \approx \frac{4}{\zeta\omega_n} \tag{7}$$

$$M_p = e^{\left(-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}\right)} \times 100\% \tag{8}$$

Substituting the obtained values:

$$t_s \approx \frac{4}{0.74 \times 15.01} \approx 0.36\,\mathrm{s} \tag{9}$$

$$M_p \approx 4.6\% \tag{10}$$

# 6 Graphical Results

### Step Response of Stabilization System

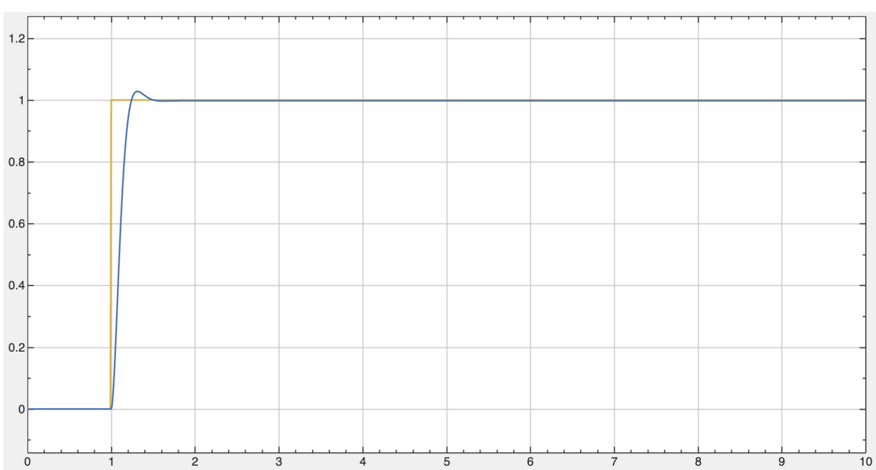


Figure 1: Closed-loop step response showing stabilization behavior

## 7 Conclusion

The project successfully demonstrates the design and implementation of a 3-axis orientation stabilization platformusing embedded control and mechatronic principles. By utilizing real-time data from the MPU6050 sensor and applying a PID-based closed-loop control system, the platform effectively reduces disturbances in roll, pitch, and yaw, maintaining stable orientation under external motion.

The system shows reliable performance in terms of fast response, minimal overshoot, and low steady-state error, validating the effectiveness of the control strategy and actuator design. MATLAB-based modeling and simulation further support the practical feasibility of the system.

Overall, the project highlights the integration of control systems, embedded programming, and mechanical design to solve real-world stabilization problems, with strong potential for applications in robotics, autonomous systems, and sensor platforms operating in dynamic environments.

## Components Used

- ESP32 Microcontroller
- MPU6050 Accelerometer and Gyroscope
- PCA9685 PWM Servo Driver
- 3x MG995 High Torque Servos
- Li-PO Battery (7.4V)
- Servo Mount Brackets
- 3D Printed Parts
- Connecting Wires
- Mounting Hardware